

Sensors and Instrumentation (EL495)

Course: Sensors and Instrumentation (EL495)

Credit Structure: (L-T-P-Cr): **3-0-2-4**

Course Placement: Technical Elective

Prerequisite Courses: Fundamentals of Electronics and Circuits, and Basic Measurement Techniques.

Instructor: Ajay Beniwal (ajay_beniwal@dau.ac.in)

Couse Content:

Unit	Content	Lectures
Unit 1	Introduction to Sensors: Sensor measurements, Sensing measurand, Sensor transfer characteristics, Sensor characteristics: Sensitivity, Linearity, Resolution, Hysteresis, Repeatability, Accuracy, Principle working of various sensors technology, Sensor design optimization methods.	10
Unit 2	Sensor's Instrumentation: Interface electronics for various sensors technology, Understanding of sensor transfer characteristics and Instrumentation design, System design for the sensors.	8
Unit 3	Sensor Fabrication Technology and Advanced Sensors: Fundamentals of fabrication processes, Contamination and clean room concepts, Wafer cleaning, Oxidation, Photolithography, Mask Design, Wet and Dry Etching, Thin Film Deposition Techniques, Advanced Sensor Concepts (Introductory Level): Chemical, MEMS, and Flexible sensors.	14
Unit 4	Intelligent Sensing and Applications: Sensor interfacing (Arduino/ESP32), IoT-enabled sensor systems, Applications: Healthcare, Agriculture & Environmental monitoring, Consumer electronics, Automotive & Defence.	10

Suggested Text Book(s)/References

1. Jacob Fraden, *Handbook of Modern Sensors: Physics, Designs and Applications*, 5th edition, Springer, 2016.
2. Tai-Ran Hsu, *MEMS & Microsystems: Design and Manufacture*, Tata McGraw-Hill, latest edition.
3. Jon S. Wilson (Ed.), *Sensor Technology Handbook*, 2nd edition, Elsevier/Academic Press, 2004.
4. S. C. Mukhopadhyay, *Intelligent Sensing, Instrumentation and Measurements*, Springer, 2013.
5. Julian W. Gardner, Florin Udrea, *Microsensors: Principles and Applications*, 2nd edition, Wiley-Blackwell, 2021.

Relevant research papers from scientific publisher like IEEE, IOP, ACS, AIP, Elsevier, RSC and others.

Course Objectives

- Understand the fundamentals of sensors, sensing measurands, and sensor characteristics.
- Learn sensor instrumentation and interface electronics for system-level design.
- Explore fabrication, advanced sensors and IoT-enabled applications.

Course Outcomes

- Identify and analyse appropriate sensors for different measurands and applications.
- Design and implement sensor systems with appropriate interfacing and measurement techniques.
- Apply knowledge of advanced sensors and intelligent sensing systems in real-world applications.

Evaluation: Mid-Sem Exam – 25%, Quiz/Assignment – 15%, End-Sem Exam – 30%, Laboratory – 30%.

Laboratory Experiment: The laboratory component includes experiments on sensor measurements, signal conditioning, sensor fabrication, testing, and IoT-enabled applications to provide hands-on experience, reinforce theoretical concepts, and develop practical skills in sensor interfacing, measurement, and intelligent sensing systems.